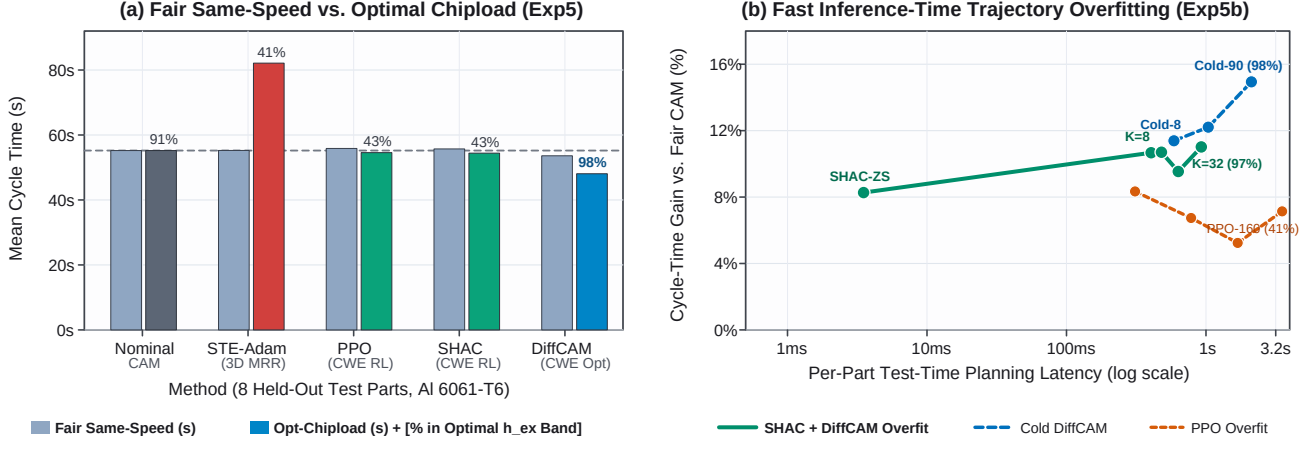
(a) Fair Same-Speed vs. Optimal Chipload (Exp5)
Mean Cycle Time (s)
80s
60s
40s
20s
0s
91%
41%
43%
43%
98%
Nominal
CAM
STE-Adam
(3D MRR)
PPO
(CWE RL)
SHAC
(CWE RL)
DiffCAM
(CWE Opt)
Method (8 Held-Out Test Parts, AI 6061-T6)
Fair Same-Speed (s)
Opt-Chipload (s) + [% in Optimal h_ex Band]
(b) Fast Inference-Time Trajectory Overfitting (Exp5b)
Cycle-Time Gain vs. Fair CAM (%)
16%
12%
8%
4%
0%
Cold-90 (98%)
Cold-8
K=8
K=32 (97%)
SHAC-ZS
PPO-160 (41%)
1ms
10ms
100ms
1s
3.2s
Per-Part Test-Time Planning Latency (log scale)
SHAC + DiffCAM Overfit
Cold DiffCAM
PPO Overfit